

30.1

Hungarian Method (1)

30.1.1 Hungarian Method: UGC NET CSE | December 2014 | Part 3 | Question: 69



Five men are available to do five different jobs. From past records, the time (in hours) that each man takes to do each job is known and is given in the following table :

		Jobs				
		I	II	III	IV	V
Men	P	2	9	2	7	1
	Q	6	8	7	6	1
	R	4	6	5	3	1
	S	4	2	7	3	1
	T	5	3	9	5	1

Find out the minimum time required to complete all the jobs.

- A. 5 B. 11 C. 13 D. 15

ugcnetcse-dec2014-paper3 assignment-problem hungarian-method

Answer key

30.2

Operation Research Pert Cpm (1)

30.2.1 Operation Research Pert Cpm: UGC NET CSE | December 2018 | Part 2 | Question: 8



In PERT/CPM, the merge event represents _____ of two or more events.

- A. completion B. beginning C. splitting D. joining

ugcnetcse-dec2018-paper2 operation-research-pert-cpm

Answer key

30.3

Simplex Method (1)

30.3.1 Simplex Method: UGC NET CSE | June 2019 | Part 2 | Question: 10



Consider an LPP given as

$$\text{Max } Z = 2x_1 - x_2 + 2x_3$$

subject to the constraints

$$\begin{aligned} 2x_1 + x_2 &\leq 10 \\ x_1 + 2x_2 - 2x_3 &\leq 20 \\ x_1 + 2x_3 &\leq 5 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

What shall be the solution of the LLP after applying first iteration of the Simplex Method?

- A. $x_1 = \frac{5}{2}, x_2 = 0, x_3 = 0, Z = 5$ B. $x_1 = 0, x_2 = 0, x_3 = \frac{5}{2}, Z = 5$
 C. $x_1 = 0, x_2 = \frac{5}{2}, x_3 = 0, Z = -\frac{5}{2}$ D. $x_1 = 0, x_2 = 0, x_3 = 10, Z = 20$

ugcnetcse-june2019-paper2 simplex-method

Answer key

Answer Keys

30.1.1

C

30.2.1

Q-Q

30.3.1

B